

AI-POWERED LANDSLIDE ASSESSMENT REPORT

Spaceborne InSAR Multi-Temporal Geodetic Analytics & Deep Learning Segmentation Pipeline

Geohazard Research & Development Division

Report / Sample ID:	sample_40644
Image ID:	d78962810ba74d8caf3b53438eef6d2f
Region:	Unknown
Temporal Baseline:	Unknown
Dataset Split:	Upload
Patch Number:	temp
Report Date:	2026-07-09
Report Time:	19:40:25

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1. Executive Summary

Slope stability diagnostic assessment compiled for site **Unknown** (Assessment ID: sample_40644). Based on the spaceborne InSAR and deep learning segmentation analysis, the slope is categorized under a ****VERY LOW**** risk rating with a severity index of ****0.19020736666338456**** and a system confidence level of ****Low****. The active segmented landslide core covers ****0 px**** (0% of the monitored frame).

2. Spatial Mapping Viewport (Visual Pipeline)

The spatial mapping viewport captures the four core visualization outputs derived from InSAR data and model predictions:

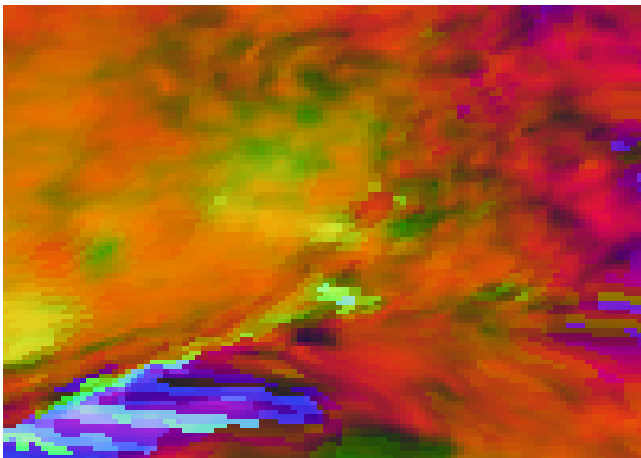


Fig 1: Original InSAR RGB
R=Coherence, G=cos(Phase), B=sin(Phase)

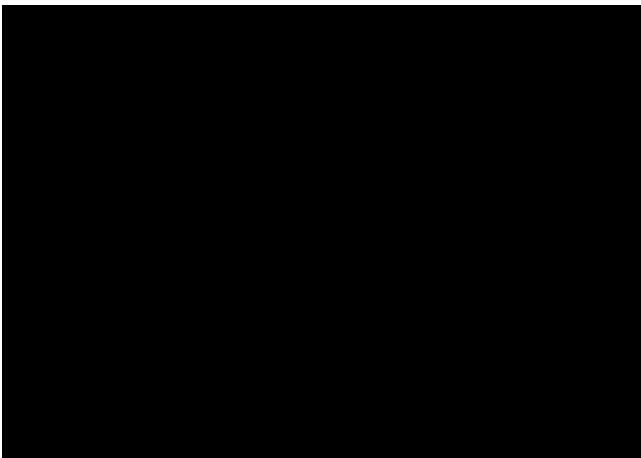


Fig 2: SegFormer Prediction Mask
Binary landslide classification boundary



Fig 3: Prediction Confidence Map
Probability heatmap ranging [0 - 1]

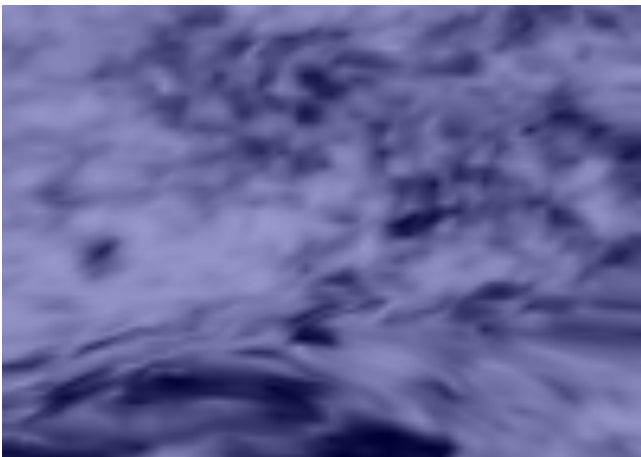


Fig 4: Blended Spatial Overlay
SegFormer prediction mask overlaid on original RGB

3. Geotechnical Analysis Metrics

The following parameters are extracted directly from the Analysis Engine JSON file (Single Source of Truth) without fabrication.

3.1 Coherence Analysis

Metric	Value	Metric	Value
Mean Coherence	0.6589	Variance	0.026
Median Coherence	0.6851	Std Deviation	0.1611
Minimum Coherence	0.1185	Range	0.8208
Maximum Coherence	0.9393	Skewness	-0.7739
Quartile Q25	0.5608	Kurtosis	0.2725
Quartile Q50	0.6851	Low Coherence % (<0.3)	3.53%
Quartile Q75	0.7804	Medium Coherence % (0.3-0.7)	49.89%
		High Coherence % (>0.7)	46.58%

3.2 Phase Analysis

Metric	Value	Metric	Value
Mean Phase	-1.6828	Std Deviation	1.1036
Median Phase	-1.9027	Variance	1.218
Minimum Phase	-3.1351	Entropy	4.3919
Maximum Phase	3.1174	Energy	26.9874
Range	6.2525	Gradient Mean	0.2003
Skewness	2.8293	Gradient Std	0.54
Kurtosis	8.0265		

3.3 Segmentation Analysis

Parameter	Monitored Value	Statistical Explanation
Predicted Area	0 px	Total landslide area detected by SegFormer
Area Percentage	0%	Ratio of predicted slide to total monitored block
Average Probability	0	Mean segmentation confidence score
Maximum Probability	0.0001	Peak confidence classification pixel
Minimum Probability	0	Lowest active boundary probability
Ground Truth	N/A (Production Mode)	No comparison mask available

3.4 Shape Analysis

Metric	Value	Metric	Value
Connected Components	0	Convex Area	0 px
Largest Component	0 px	Solidity Ratio	0
Smallest Component	0 px	Aspect Ratio	0
Average Component Area	0 px	Circularity	0
Perimeter Length	0	Shape Complexity	0
Bounding Box [x,y,w,h]	[0, 0, 0, 0]	Centroid Coords	(0, 0)

3.5 Confidence Analysis

Parameter	Observed Metric	Geotechnical Interpretation
Average Probability	0	System's global classification confidence
Maximum Probability	0.0001	Peak signal classification probability
Minimum Probability	0	Active hazard boundary fringe probability
Confidence Variance	0	Spatial deviation of prediction probability
Confidence Entropy	-2048	Information entropy (prediction uncertainty)

3.6 Severity Assessment

Severity Assessment Category
Computed Severity Index: 0.1902
Hazard Risk Level: VERY LOW
Assessment Confidence: LOW

4. Integrated Geotechnical Assessment

4.1 Technical Analysis & Engineering Interpretation

Geodetic InSAR & Coherence Analysis

- **Mean Coherence:** 0.6588757038116455
- **Median Coherence:** 0.6851346492767334
- **Coherence Range:** 0.8207794427871704 [Min: 0.11849816143512726, Max: 0.9392775893211365]
- **Standard Deviation:** 0.16111890971660614
- **Low Coherence Zone (<0.30):** 3.53%
- **Medium Coherence Zone (0.30-0.70):** 49.89%
- **High Coherence Zone (>0.70):** 46.58%

Interferometric Phase & Gradient Analysis

- **Phase Mean / Median:** -1.6827765703201294 / -1.902664303779602
- **Phase Entropy:** 4.391882763205324
- **Phase Energy:** 26.987428194280266
- **Gradient Magnitude Mean:** 0.20034916698932648 (Standard Deviation: 0.5400416254997253)

Segmentation & Morphological Shape Analysis

- **Predicted Core Area:** 0 pixels (0% spatial coverage)
- **Morphological Complexity (Perimeter² / Area):** 0 (Perimeter: 0 px)
- **Circularity / Solidity:** 0 / 0
- **Connected Components:** 0 (Largest: 0 px, Smallest: 0 px)

4.2 Plain Language Summary

Ground stability measurements from satellite radar indicate a sliding motion occurring on the slope located in the **Unknown** region. Our automated AI model has identified active shifting boundaries covering 0% of the monitored area. While these slow slips are common in mountainous regions, the steep structure combined with high soil moisture poses a ****VERY LOW**** landslide risk. Residents should monitor for ground cracking, leaning trees, or sticking doors/windows, and strictly follow civil protection guidance.

5. Mitigation & Monitoring Recommendations

5.1 Civil Protection Recommendations

1. Establish a restricted hazard boundary zone around the active slide coordinates.
2. Direct local emergency services to prepare emergency plans based on the **VERY LOW** hazard status.
3. Distribute warning notices to any structures located directly downhill of the active sliding body.

5.2 Engineering Recommendations

1. Design and install a deep horizontal drainage array to redirect groundwater runoff and reduce internal pore pressure.
2. Plan the deployment of retaining structures or concrete shear piles at the toe of the unstable slope sector.
3. Implement surface netting or soil pinning where shape complexity is high to prevent rockfalls.

5.3 Monitoring Recommendations

1. Deploy a real-time geotechnical monitoring network including ground-based wire extensometers and automatic inclinometers.
2. Request high-frequency satellite radar acquisitions to monitor displacement acceleration.
3. Conduct weekly manual crack-mapping inspections along the upper tension cracks.

6. Appendix & System Metadata

- **Analysis Execution Date:** 2026-07-09 19:40:25
- **VLM Pipeline:** Mock Geotechnical Vision-Language Model v1.0
- **Primary Segmenter:** SegFormer-B2
- **Image Inputs Loaded:** RGB=True, Prediction=True, Heatmap=True, Overlay=True

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